Notes: Abowd, Kramarz, and Margolis (Econometrica, 1999)

High Wage Workers and High Wage Firms

Contents

1 Big picture

- **Question.** How much of wage variation is due to workers, how much is due to firms, and how do worker and firm heterogeneity explain classic wage differentials such as industry wage effects and firm-size wage effects?
- **Main data contribution.** The paper uses a massive matched employer–employee data set from France, linking more than one million workers to more than five hundred thousand employing firms over time.
- **Core empirical object.** The authors decompose log real annual compensation into observable worker characteristics, person effects, firm effects, and residual variation.
- **Central conceptual distinction.** A high-wage worker is a worker with compensation above what observables predict; a high-wage firm is a firm that pays above what worker observables and worker heterogeneity predict.
- **Main result at worker level.** Person effects are the dominant non-residual source of wage variation. The unobserved person component is especially important.
- **Main result at firm level.** Firm effects are economically important but less important than person effects in explaining individual compensation.
- **Assortative matching result.** High-wage workers tend to be employed by higher-wage firms, but the estimated correlation between person effects and firm effects is not large.
- **Industry wage differentials.** Person effects explain about 90% of French inter-industry wage differentials, while firm effects explain much less.
- **Firm-size wage effects.** Person effects explain about 75% of the firm-size wage premium, while firm effects again explain a smaller component.
- **Firm-level performance result.** Firms that employ high-wage workers are more productive but not necessarily more profitable. Firms that pay higher wages controlling for worker effects are more productive and more profitable.
- **Methodological contribution.** The paper develops practical estimators for enormous two-way fixed-effect wage models before full brute-force computation of the entire least-squares system was feasible.
- **Takeaway.** The wage structure cannot be understood from either worker data or firm data alone. Matched employer–employee data reveal that worker heterogeneity, firm wage policies, and sorting jointly shape observed wage differentials.

2 Data and measurement

2.1 French matched employer–employee data

- The analysis uses the *Declarations annuelles des salaires* (DAS), a French administrative employer declaration system.
- The DAS records annual total compensation paid by firms to workers.
- The matched data connect individual workers to their employing firms, allowing worker histories and firm employment relationships to be followed over time.
- The core estimation sample covers private-sector French workers and firms over 1976–1987.
- The data include more than one million workers and hundreds of thousands of firms.

Interpretation: The key advantage is simultaneous observation of worker and firm identities across years. This permits decomposition of wages into worker and firm components and uses job mobility to identify firm effects separately from worker effects.

2.2 Dependent variable: real annual compensation

The main dependent variable is log real total annual compensation:

$$y_{it} = \log(\text{real annual compensation}_{it}). \tag{1}$$

- Compensation is annual and includes total labor compensation paid by the employer.
- Compensation is deflated to real units.
- The main analysis uses log compensation rather than hourly wages.

Interpretation: The use of annual compensation means that the estimated effects combine wage rates and annual labor supply conditional on employment. The paper is therefore best interpreted as a decomposition of annual labor compensation.

2.3 Worker observables

The worker-level observables include:

- labor-force experience;
- powers of experience;
- education categories;
- region indicators;
- year indicators;
- seniority with the current firm;
- sex-specific estimations.

Interpretation: Observable characteristics explain part of compensation, but the main contribution is estimating the much larger unobserved person and firm components.

2.4 Firm-level data

The paper also constructs firm-level variables, including:

- employment;
- real total assets;
- real value added;
- operating income per unit of capital;
- capital per employee;
- occupational composition of the workforce;
- firm age and firm survival.

Interpretation: These variables allow the authors to ask whether firms with high wage policies or high-wage workers are more productive, more profitable, more capital intensive, or more skill intensive.

2.5 Mobility and identification

- Identification of both person and firm effects requires workers to move across firms.
- Workers who never move identify person effects but do not directly connect firms.
- Movers connect employers and allow firm effects to be compared.
- Table I documents that many workers have only one employer, but a large share work in firms that are connected through movers.

Interpretation: The connectedness of the worker–firm mobility network is the foundation of the AKM design. Without movers, firm effects and worker effects would be observationally confounded.

2.6 Why the French setting matters

- The dataset is administrative and very large.
- The French labor market has strong institutions and wage structures that make industry, firm-size, and firm-policy questions especially interesting.
- Matched longitudinal data allow the paper to revisit classic questions such as inter-industry wage differentials with much better controls.

Interpretation: The contribution is not only French-specific. The paper created the empirical template now known as the AKM wage decomposition.

3 Statistical model and empirical specifications

3.1 General compensation equation

The core wage equation is a two-way worker–firm effects model:

$$y_{it} = x_{it}'\beta + \theta_i + \psi_{J(i,t)} + \varepsilon_{it}, \tag{2}$$

where:

- y_{it} is log real annual compensation for worker i in year t;
- x_{it} is a vector of time-varying worker characteristics;
- θ_i is a worker/person effect;
- $\psi_{J(i,t)}$ is the wage policy effect of the firm employing worker i in year t;
- ε_{it} is residual variation.

Interpretation: This equation is the central AKM decomposition. It separates compensation into what can be predicted from observed worker characteristics, who the worker is, and which firm employs the worker.

3.2 Person effects

The person effect is decomposed into an observable time-invariant part and an unobservable part:

$$\theta_i = \alpha_i + \eta_i'\mu, \tag{3}$$

where η_i includes time-invariant measured personal characteristics such as education and α_i is the unobserved person component.

Interpretation: The paper distinguishes observed human capital, such as education, from unobserved worker heterogeneity. The latter becomes the dominant component of compensation variation.

3.3 Firm effects with seniority

The firm effect is allowed to vary with seniority. A linear version is:

$$\psi_{J(i,t)} = \phi_j + \gamma_j s_{it}, \tag{4}$$

where s_{it} is seniority of worker i in firm j at time t.

The richer version allows a change in slope after ten years:

$$\psi_{J(i,t)} = \phi_j + \gamma_j s_{it} + \gamma_{2j} T_1(s_{it} - 10), \tag{5}$$

where $T_1(x) = 0$ if $x < 0$ and $T_1(x) = x$ if $x > 0$.

Interpretation: Firms may differ not only in wage intercepts but also in their internal wage growth and seniority policies. This lets the paper analyze internal wage structures, not just firm wage premia.

3.4 Matrix notation

The model can be written as:

$$y = X\beta + D\theta + F\psi + \varepsilon. \tag{6}$$

Here D maps observations to workers and F maps observations to firms.

Interpretation: The matrix notation makes clear why the problem is computationally hard: D has roughly one column per worker and F has roughly one column per firm.

3.5 Identification problem

- Person and firm effects are separately identified only within connected mobility groups.
- If a worker never moves and a firm has no connections to movers, the worker and firm effects cannot be separately distinguished.
- Identification comes from comparing the same worker across firms and different workers within firms.

Interpretation: The model is conceptually a linear regression, but the relevant design matrix is enormous and sparse. This makes ordinary least squares computationally infeasible for the full problem in the paper's era.

3.6 Conditional estimation methods

The paper develops several estimation methods:

- **Order-independent conditional estimation:** conditions on selected functions of firm and worker design matrices to estimate worker and firm components.
- **Order-dependent persons-first estimation:** estimates person effects first and then firm effects.
- **Order-dependent firms-first estimation:** estimates firm effects first and then person effects.
- **Consistent estimation on subsets:** uses smaller problems, especially the largest firms, where more exact least squares comparisons are computationally feasible.

Interpretation: Because full least squares is too large, the authors use alternative estimators and compare their implications. A key lesson is that persons-first methods appear more reliable for person effects.

3.7 Decomposition of classic wage differentials

For an industry or firm-size group g, the raw wage effect can be decomposed into group averages of worker and firm components:

$$WageEffect_g \approx \overline{x\beta}_g + \overline{\alpha}_g + \overline{\eta\mu}_g + \overline{\psi}_g. \tag{7}$$

Interpretation: This asks whether high-wage industries or large firms pay more because they employ better workers, because they have high firm wage policies, or both.

3.8 Firm-level outcome regressions

At the firm level, the paper regresses productivity, profitability, factor use, and survival on the firm averages of wage components:

$$Outcome_j = a + b_1\overline{x\beta}_j + b_2\overline{\alpha}_j + b_3\overline{\eta\mu}_j + b_4\psi_j + Controls_j + u_j. \tag{8}$$

Interpretation: These regressions ask what high-wage workers and high-wage firms imply for firm performance. The distinction between worker quality and firm wage policy is critical.

4 Identification and empirical design

4.1 What identifies person effects

- Repeated observations on the same worker identify persistent person-specific wage components.
- Workers observed across years allow the model to separate permanent worker heterogeneity from time-varying observables.
- Movers help distinguish worker effects from firm effects.

Interpretation: Person effects represent persistent wage differences among workers that are not captured by observed characteristics.

4.2 What identifies firm effects

- Firm effects are identified by movements of workers between firms and by comparisons among workers in connected firms.
- A firm premium is meaningful only relative to other firms in the same connected set.
- The paper's data have sufficient mobility to estimate firm effects for a large share of the sample.

Interpretation: The firm effect is not just the average wage at the firm. It is the wage premium after controlling for observed and unobserved worker heterogeneity.

4.3 Why ordinary wage regressions are insufficient

- A regression of wages on industry or firm size confounds worker quality with firm wage policies.
- High-wage industries may employ high-wage workers rather than paying high firm premia.
- Large firms may pay more because they select high-wage workers.

Interpretation: The AKM decomposition turns wage differentials into a sorting question: are high wages due to firms, workers, or worker–firm allocation?

4.4 Specification checks

- The paper compares multiple estimation methods.
- It studies correlations among components generated by different methods.
- It compares conditional estimates to full least-squares estimates on the 115 largest firms.
- It examines variance decompositions and correlations among wage components.

Interpretation: The paper's preferred conclusions rely not on one estimator but on consistency across diagnostics: person effects are robustly important; firm effects matter but are smaller.

4.5 Limits of the design

- The data measure annual compensation, not necessarily hourly wages.
- Some person and firm effects are estimated with error, especially when workers or firms have few observations.
- Full two-way least squares is computationally infeasible for the full sample in the paper.

- Mobility may be nonrandom, and the linear fixed-effects model abstracts from match-specific effects.
- The sample is French private-sector employment, so institutional context matters.

Interpretation: The paper is best understood as a foundational decomposition of wage variance and wage differentials, not as a causal model of why workers and firms match.

5 Tables and figures

5.1 Table I: mobility structure and identification

Read:

- Table I reports the structure of individual histories by number of sample years and number of employers.
- About 59.4% of sample individuals never change employers.
- A large fraction of single-employer workers nevertheless work in firms that employ movers.
- The existence of movers connects workers and firms and makes estimation of firm effects possible.

Economic message: Mobility is the empirical source of identification. Without it, worker and firm effects would be inseparable.

TABLE I

STRUCTURE OF THE INDIVIDUAL DATA BY YEARS IN SAMPLE AND NUMBER OF EMPLOYERS
(Number of Individuals, Most Common Configuration of Employers)

Years in Sample	Number of Employers 1	1a	2	3	4	5	6	7	8	9	10	Total	Percent
1	318,627	247,532										318,627	27.3%
	1	1											
2	75,299	57,411	51,066									126,365	10.8%
	2	2	11										
3	46,385	36,540	32,947	19,583								98,915	8.5%
	3	3	12	111									
4	43,019	34,922	26,631	17,191	8,330							95,171	8.2%
	4	4	13	112	1111								
5	41,130	34,596	26,408	15,291	8,685	3,610						95,124	8.2%
	5	5	14	113	1112	11111							
6	29,755	25,388	20,953	13,734	7,592	4,073	1,653					77,760	6.7%
	6	6	15	114	1113	11112	111111						
7	19,413	16,709	17,384	12,039	7,305	3,864	1,931	735				62,671	5.4%
	7	7	16	115	1114	11113	111112	1111111					
8	23,484	20,378	20,421	13,185	7,673	4,001	2,061	917	327			72,069	6.2%
	8	8	44	116	1115	11114	111113	1111112	11111111				
9	38,505	34,147	26,350	15,791	8,590	4,383	2,104	938	362	114		97,137	8.3%
	9	9	54	117	1116	11115	111114	1111113	11111112	111111111			
10	56,881	51,425	32,616	17,728	8,369	3,839	1,837	739	314	109	34	122,466	10.5%
	10[a]	10[a]	64	118	1117	11116	221113	1131112	11111113	111111112	1111111111		
Total	692,498	559,048	254,776	124,542	56,544	23,770	9,586	3,329	1,003	223	34	1,166,305	100.0%
Percent	59.4%	47.9%	21.8%	10.7%	4.8%	2.0%	0.8%	0.3%	0.1%	0.0%	0.0%	100.0%	

Notes: Employment configurations are described in terms of the number of consecutive years spent with each of the individual's employers, in order (e.g. configuration 124 means that the individual spent 1 year with his first employer, then 2 years with his second employer, and finally 4 years with his third employer). Column 1a refers to the subset of individuals with only one employer whose employing firm had at least one other individual who had changed firms at least once in his career (required for least squares identification of both firm and individual effects). (a) This configuration corresponds to 10 years of data with the first (and only) employer.
Source: Authors' calculations based on the Déclarations Annuelles des Salaires (DAS).

TABLE II

COMPARISON OF LEAST SQUARES ESTIMATES OF WAGE DETERMINATION IN FRANCE AND THE UNITED STATES 1982–1987

	France				*United States*			
	Men		Women		Men		Women	
Variable	Mean	OLS Results	Mean	OLS Results	Mean	OLS Results	Mean	OLS Results
Intercept	1.000	1.365	1.000	1.163	1.000	0.534	1.000	0.380
	(0.000)	(6.746E-3)	(0.000)	(8.190E-3)	(0.000)	(5.614E-3)	(0.000)	(5.679E-3)
Years of	10.726	0.077	11.325	0.088	11.880	0.061	12.300	0.072
Education	(3.659)	(2.848E-4)	(3.267)	(3.998E-4)	(2.391)	(3.521E-4)	(2.149)	(3.712E-4)
Experience	20.722	0.058	19.048	0.060	15.894	0.112	16.036	0.082
	(12.222)	(1.228E-3)	(12.163)	(1.435E-3)	(12.311)	(9.219E-4)	(12.323)	(8.899E-4)
Experience^2/	5.788	−0.104	5.108	−0.188	4.042	−0.506	4.090	−0.436
100	(5.875)	(9.095E-3)	(5.533)	(1.133E-2)	(5.251)	(8.487E-3)	(5.077)	(8.482E-3)
Experience^3/	18.915	−0.007	16.207	0.018	12.611	0.102	12.544	0.093
1,000	(26.344)	(2.554E-3)	(23.730)	(3.355E-3)	(21.849)	(2.811E-3)	(20.514)	(2.898E-3)
Experience^4/	68.009	0.002	56.700	0.001	43.914	−0.007	42.506	−0.007
10,000	(120.682)	(2.397E-4)	(103.687)	(3.316E-4)	(92.920)	(3.037E-4)	(84.740)	(3.232E-4)
1982	0.175	0.036	0.167	0.027	0.163	0.072	0.160	0.019
	(0.380)	(3.001E-3)	(0.373)	(3.784E-3)	(0.370)	(2.715E-3)	(36.707)	(2.596E-3)
1983	0.170	0.018	0.166	0.006	0.162	0.049	0.160	0.015
	(0.375)	(3.020E-3)	(0.373)	(3.783E-3)	(0.369)	(2.707E-3)	(0.367)	(2.579E-3)
1984	0.166	0.019	0.164	0.012	0.164	0.032	0.162	0.000
	(0.372)	(3.033E-3)	(0.371)	(3.793E-3)	(0.370)	(2.679E-3)	(0.368)	(2.557E-3)
1985	0.165	0.005	0.165	−0.001	0.167	0.018	0.166	−0.002
	(0.371)	(3.040E-3)	(0.371)	(3.785E-3)	(0.373)	(2.658E-3)	(0.372)	(2.534E-3)
1986	0.162	0.023	0.168	0.018	0.174	0.015	0.175	0.004
	(0.369)	(3.051E-3)	(0.374)	(3.767E-3)	(0.379)	(2.601E-3)	(0.380)	(2.479E-3)
Paris Region	0.210	0.168	0.240	0.158	—	—	—	—
	(0.407)	(2.147E-3)	(0.427)	(2.567E-3)	—	—	—	—
Northeast U.S.	—	—	—	—	0.210	−0.046	0.217	−0.057
	—	—	—	—	(0.408)	(2.496E-3)	(0.412)	(2.393E-3)
Midwest U.S.	—	—	—	—	0.263	−0.039	0.273	−0.088
	—	—	—	—	(0.440)	(2.309E-3)	(0.446)	(2.222E-3)
Southern U.S.	—	—	—	—	0.296	−0.143	0.289	−0.128
	—	—	—	—	(0.457)	(2.206E-3)	(0.453)	(2.151E-3)
Observations	165,036	165,036	126,320	126,320	259,297	259,297	259,266	259,266
Adjusted R^2		0.3866		0.3254		0.3626		0.2428

Sources: Enquête Emploi (1982–1987) for France and NBER outgoing rotation group CPS extracts (1982–1987) for the United States. *Notes:* Standard Deviations/Errors in Parentheses. Both regressions included only individuals between 16 and 60 years old, inclusive. Both regressions used the sample weights. Experience is measured as (age) − (age at the end of schooling) in France and (age) − (years of schooling) − 5 in the United States.

runs from 1976 through 1987, with 1981 and 1983 excluded because the underlying administrative data were not sampled in those years. The initial data set contained 7,416,422 observations. Each observation corresponded to a unique individual-year-establishment combination. An observation in this initial DAS

5.2 Table II: French and U.S. wage-determination comparison

Read:

- Table II compares least-squares estimates for France and the United States over 1982–1987.
- The table reports means and OLS coefficients for variables such as education, experience, region, year effects, and industry.
- The estimates provide context for the French sample and for standard wage regressions that do not include person and firm effects.

Economic message: Standard wage regressions can describe raw wage differences but cannot reveal whether high wages are due to workers or firms.

5.3 Table III: compensation equation estimates

Read:

- Table III summarizes estimates of the general compensation equation under several estimation methods.
- It includes consistent methods, conditional methods, least squares without person/firm effects, within-person estimators, within-firm estimators, and limited-firm-effect methods.
- Comparing these columns shows how including person and firm effects changes the coefficients on observed characteristics.
- The table also reports specification test statistics.

Economic message: The inclusion of person and firm effects materially changes wage-regression interpretation. The table demonstrates why standard wage equations are incomplete.

TABLE III

ESTIMATES OF THE EFFECTS OF LABOR FORCE EXPERIENCE, REGION, YEAR, EDUCATION, INDIVIDUALS, AND FIRMS ON THE LOG OF REAL TOTAL ANNUAL COMPENSATION COSTS INDIVIDUAL DATA BY SEX FOR 1976 TO 1987

Variable	*Consistent Method Person & Firm Effects* Parameter Estimate	Standard Error	*Conditional Method Persons First* Parameter Estimate	Standard Error	*Least Squares No Person / Firm Effects* Parameter Estimate	Standard Error	*Within Persons No Firm Effects* Parameter Estimate	Standard Error	*Within Persons Limited Firm Effects*[a] Parameter Estimate	Standard Error	*Within Firms No Person Effects* Parameter Estimate	Standard Error	*Within Industry No Person Effects* Parameter Estimate	Standard Error	*Within Firm Size No Person Effects* Parameter Estimate	Standard Error
Men																
Total Labor Force Experience	0.0586	(0.0015)	0.0687	(0.0004)	0.0542	(0.0003)	0.0695	(0.0004)	0.0685	(0.0004)	0.0448	(0.0003)	0.0521	(0.0003)	0.0507	(0.0003)
(Labor Force Experience)^2/100	−0.3432	(0.0072)	−0.4415	(0.0027)	−0.2280	(0.0030)	−0.4543	(0.0029)	−0.4446	(0.0030)	−0.1584	(0.0027)	−0.2115	(0.0030)	−0.2047	(0.0029)
(Labor Force Experience)^3/1,000	0.0734	(0.0025)	0.1053	(0.0010)	0.0503	(0.0010)	0.1100	(0.0010)	0.1074	(0.0010)	0.0298	(0.0009)	0.0452	(0.0010)	0.0441	(0.0010)
(Labor Force Experience)^4/10,000	−0.0057	(00003)	−0.0093	(0.0001)	−0.0046	(0.0001)	−0.0099	(0.0001)	−0.0096	(0.0001)	−0.0025	(0.0001)	−0.0041	(0.0001)	−0.0040	(0.0001)
Lives in Ile-de-France	0.0051	(0.0017)	0.0832	(0.0010)	0.1398	(0.0005)	0.0819	(0.0011)	0.0805	(0.0011)	0.1117	(0.0007)	0.1316	(0.0006)	0.1314	(0.0005)
Year 1977	0.0245	(0.0009)	0.0251	(0.0007)	0.0343	(0.0010)	0.0252	(0.0007)	0.0248	(0.0007)	0.0182	(0.0009)	0.0275	(0.0010)	0.0296	(0.0010)
Year 1978	0.0643	(0.0017)	0.0605	(0.0008)	0.0645	(0.0010)	0.0609	(0.0008)	0.0598	(0.0008)	0.0463	(0.0009)	0.0560	(0.0010)	0.0581	(0.0010)
Year 1979	0.0887	(0.0024)	0.0879	(0.0009)	0.0841	(0.0010)	0.0883	(0.0010)	0.0873	(0.0010)	0.0598	(0.0009)	0.0755	(0.0010)	0.0774	(0.0010)
Year 1980	0.1081	(0.0031)	0.1030	(0.0011)	0.0899	(0.0010)	0.1033	(0.0012)	0.1024	(0.0012)	0.0644	(0.0009)	0.0803	(0.0010)	0.0841	(0.0010)
Year 1982	0.1473	(0.0035)	0.1441	(0.0014)	0.1137	(0.0010)	0.1447	(0.0016)	0.1434	(0.0016)	0.0809	(0.0009)	0.1043	(0.0010)	0.1091	(0.0010)
Year 1984	0.1872	(0.0041)	0.1911	(0.0018)	0.1441	(0.0010)	0.1919	(0.0020)	0.1903	(0.0020)	0.1009	(0.0009)	0.1316	(0.0011)	0.1386	(0.0011)
Year 1985	0.2044	(0.0047)	0.2173	(0.0020)	0.1662	(0.0011)	0.2179	(0.0022)	0.2162	(0.0022)	0.1146	(0.0009)	0.1516	(0.0011)	0.1612	(0.0011)
Year 1986	0.2366	(0.0053)	0.2529	(0.0022)	0.1841	(0.0010)	0.2535	(0.0024)	0.2517	(0.0024)	0.1315	(0.0009)	0.1690	(0.0011)	0.1813	(0.0011)
Year 1987	0.2499	(0.0060)	0.2763	(0.0024)	0.1954	(0.0010)	0.2768	(0.0026)	0.2749	(0.0026)	0.1401	(0.0009)	0.1808	(0.0011)	0.1948	(0.0011)
Women																
Total Labor Force Experience	0.0144	(0.0016)	0.0290	(0.0005)	0.0326	(0.0004)	0.0308	(0.0006)	0.0298	(0.0005)	0.0224	(0.0004)	0.0603	(0.0004)	0.0286	(0.0004)
(Labor Force Experience)^2/100	−0.1063	(0.0091)	−0.1728	(0.0036)	−0.1117	(0.0038)	−0.1771	(0.0041)	−0.1729	(0.0040)	−0.0318	(0.0035)	−0.3525	(0.0039)	−0.0816	(0.0038)

TABLE IV

DESCRIPTIVE STATISTICS FOR COMPONENTS OF LOG REAL TOTAL ANNUAL COMPENSATION BY SEX FOR 1976 TO 1987

Variable Definition	*Men* Mean	Std. Dev.	*Women* Mean	Std. Dev.
Log (Real Annual Compensation, 1980 FF)	4.3442	0.5187	4.0984	0.4801
Order-Independent Method				
$x\beta$, Predicted Effect of x Variables	0.3890	0.1489	0.2849	0.1144
θ, Individual Effect Including Education	3.9552	0.4475	3.8135	0.3930
α, Individual Effect (Unobserved Factors)	0.0000	0.4051	0.0000	0.3771
$u\eta$, Individual Effect of Education and Sex	3.9552	0.1902	3.6893	0.1107
ψ, Firm Effect (Intercept and Slope)	−0.0363	0.4642	0.0665	0.5116
ϕ, Firm Effect Intercept	−0.1367	0.4532	−0.0235	0.4967
γ, Firm Effect Slope	0.0149	0.0503	0.0172	0.0531
Order-Dependent Method: Persons First				
$x\beta$, Predicted Effect of x Variables	0.4261	0.1383	0.3234	0.1120
θ, Individual Effect Including Education	3.9160	0.4387	3.7776	0.3843
α, Individual Effect (Unobserved Factors)	0.0000	0.3947	0.0000	0.3639
$u\eta$, Individual Effect of Education and Sex	3.9160	0.1915	3.7776	0.1238
ψ, Firm Effect (Intercept and Slope)	0.0028	0.0685	−0.0039	0.0566
ϕ, Firm Effect Intercept	0.0031	0.1044	−0.0072	0.0969
γ, Firm Effect Slope	−3.37e-05	0.0335	8.28e-04	0.0326
γ_2, Firm Effect Slope Change at 10 years	−5.36e-04	0.0542	−1.64e-03	0.0574
Other Estimation Methods				
Seniority coefficient, Least Squares (Standard Error)	0.0118	(0.0001)	0.0141	(0.0001)
Seniority coefficient, within Persons (Standard Error)	0.0033	(0.0001)	0.0024	(0.0001)
Seniority coefficient, within Firms (Standard Error)	0.0078	(0.0001)	0.0102	(0.0001)
Seniority coefficient, within Industry (Standard Error)	0.0090	(0.0001)	0.0121	(0.0001)
Seniority coefficient, within Size Class (Standard Error)	0.0097	(0.0001)	0.0126	(0.0001)

5.4 Tables IV–VI: compensation components and diagnostics

Read:

- Table IV reports descriptive statistics for components of log real annual compensation.
- The unobserved individual effect has a large standard deviation relative to other components.
- Firm effects are important but less dispersed than person effects.
- Table V compares correlations among components generated by different estimation methods.
- Table VI decomposes variance and correlations among wage components under the preferred methods.

Economic message: The diagnostic tables establish the central empirical result: persistent worker heterogeneity is a much larger source of wage variation than firm heterogeneity.

TABLE V

CORRELATIONS AMONG THE COMPONENTS OF PERSON AND FIRM HETEROGENEITY AS ESTIMATED BY THE ORDER-INDEPENDENT, ORDER-DEPENDENT, FULL LEAST SQUARES ON THE 115 LARGEST FIRMS, AND CONSISTENT METHODS

Source of Estimate of the Indicated Effect	Parameter Name	Simple Correlation with: Order-Independent Estimates		Order-Dependent Estimates			Full Least Squares Estimates on the 115 Largest Firms		Consistent Estimates
		Persons First							
Firm Effects		ϕ	γ	ϕ	γ	γ_2	ϕ	γ	γ
Order-Independent Estimates	ϕ	1.0000	−0.0718	0.1553	−0.0837	0.0188	0.0888	0.2800	0.0361
	γ	−0.0718	1.0000	−0.2202	0.5300	−0.0077	−0.3276	0.3126	0.0907
Order-Dependent Estimates (Persons First)	ϕ	0.1553	−0.2202	1.0000	−0.5625	0.2562	0.6659	−0.0231	−0.1810
	γ	−0.0837	0.5300	−0.5625	1.0000	−0.2094	−0.6580	0.2739	0.1358
	γ_2	0.0188	−0.0077	0.2562	−0.2094	1.0000	0.5492	0.0293	−0.0126
Full Least Squares Estimates Using the 115 Largest Firms	ϕ	0.0888	−0.3276	0.6659	−0.6580	0.5492	1.0000	−0.1841	−0.1964
	γ	0.2800	0.3126	−0.0231	0.2739	0.0293	−0.1841	1.0000	0.5106
Consistent Estimates	γ	0.0361	0.0907	−0.1810	0.1358	−0.0126	−0.1964	0.5106	1.0000
		Firms First							
Person Effects		α		α			α		
Order-Independent Estimates	α	1.0000		0.5833			0.9896		
Order-Dependent Estimates (Firms First)	α	0.5833		1.0000			0.5983		
Full Least Squares Estimates Using the 115 Largest Firms	α	0.9896		0.5983			1.0000		

TABLE VI

SUMMARY STATISTICS FOR THE DECOMPOSITION OF VARIANCE USING THE ORDER-INDEPENDENT AND THE ORDER-DEPENDENT CONDITIONAL METHODS FOR INDIVIDUAL DATA, BOTH SEXES, 1976–1987

Order-Independent Estimation Variable Description	Mean	Std. Dev.	Simple Correlation with: y	$x\beta$	θ	α	$u\eta$	ψ	ϕ	$s\gamma$	γ
y, Log (Real Annual Compensation, 1980 FF)	4.2575	0.5189	1.0000	0.2614	0.8962	0.8015	0.4011	0.2604	0.1603	0.2729	0.0333
$x\beta$, Predicted Effect of x Variables	0.3523	0.1464	0.2614	1.0000	−0.0445	−0.1243	0.1509	0.0697	0.0824	−0.0279	0.0300
θ, Individual Effect Including Education[a]	3.9052	0.4335	0.8962	−0.0445	1.0000	0.8964	0.4433	0.2965	0.1717	0.3384	0.0387
α, Individual Effect (Unobserved Factors)[a]	0.0000	0.3955	0.8015	−0.1243	0.8964	1.0000	0.0000	0.2640	0.1465	0.3178	0.0372
$u\eta$, Individual Effect of Education	3.9052	0.1776	0.4011	0.1509	0.4433	0.0000	1.0000	0.1349	0.0910	0.1209	0.0122
ψ, Firm Effect (Intercept and Slope)	0.0000	0.4839	0.2604	0.0697	0.2965	0.2640	0.1349	1.0000	0.9259	0.2537	0.0860
ϕ, Firm Effect Intercept	−0.0968	0.4721	0.1603	0.0824	0.1717	0.1465	0.0910	0.9259	1.0000	−0.1305	−0.0718
$s\gamma$, Firm Effect of Seniority	0.0968	0.1844	0.2729	−0.0279	0.3384	0.3178	0.1209	0.2537	−0.1305	1.0000	0.4094
γ, Firm Effect Slope	0.0157	0.0513	0.0333	0.0300	0.0387	0.0372	0.0122	0.0860	−0.0718	0.4094	1.0000

Order-Dependent Estimation: Persons First Variable Description	Mean	Std. Dev.	Simple Correlation with: y	$x\beta$	θ	α	$u\eta$	ψ	ϕ	$s\gamma + T(s)\gamma_2$	γ	γ_2
y, Log (Real Annual Compensation, 1980 FF)	4.2575	0.5189	1.0000	0.3271	0.9310	0.7331	0.4143	0.2131	0.1303	0.0053	0.0293	0.0276
$x\beta$, Predicted Effect of x Variables	0.3899	0.1386	0.3271	1.0000	0.0787	−0.0290	0.2211	0.0325	0.0350	−0.0157	−0.0148	0.0077
θ, Individual Effect Including Education[a]	3.8672	0.4255	0.9310	0.0787	1.0000	0.8842	0.4769	0.1079	0.0889	−0.0223	−0.0190	0.0225
α, Individual Effect (Unobserved Factors)[a]	0.0000	0.3841	0.7331	−0.0290	0.8842	1.0000	0.0000	0.0926	0.0828	−0.0263	−0.0202	0.0202
$u\eta$, Individual Effect of Education	3.8672	0.1831	0.4143	0.2211	0.4769	0.0000	1.0000	0.0473	0.0263	0.0041	−0.0006	0.0081
ψ, Firm Effect (Intercept and Slope)	0.0000	0.0647	0.2131	0.0325	0.1079	0.0926	0.0473	1.0000	0.4428	0.2089	−0.0909	0.0717
ϕ, Firm Effect Intercept	−0.0009	0.1019	0.1303	0.0350	0.0889	0.0828	0.0263	0.4428	1.0000	−0.7844	−0.5625	0.2562
$s\gamma + T(s)\gamma_2$, Firm Effect of Seniority	0.0009	0.0935	0.0053	−0.0157	−0.0223	−0.0263	0.0041	0.2089	−0.7844	1.0000	0.5507	−0.2298
γ, Firm Effect Slope	0.0003	0.0332	−0.0293	−0.0148	−0.0190	−0.0202	−0.0006	−0.0909	−0.5625	0.5507	1.0000	−0.2094
γ_2, Change in Firm Effect Slope	−0.0009	0.0553	0.0276	0.0077	0.0225	0.0202	0.0081	0.0717	0.2562	−0.2298	−0.2094	1.0000

Notes: (a) Correlations have been corrected for the sampling variance of the estimated effect.

5.5 Tables VII–VIII and Figures 1–3: industry and firm-size wage effects

TABLE VII

GENERALIZED LEAST SQUARES ESTIMATION BETWEEN INDUSTRY WAGE EFFECTS AND INDUSTRY AVERAGES OF FIRM-SPECIFIC COMPENSATION POLICIES

Independent Variable	Coefficient	Standard Error	Coefficient	Standard Error	Coefficient	Standard Error
	Based on Order-Independent Estimates					
Industry Average α	1.0390	(0.0023)	1.0053	(0.0022)		
Industry Average ψ	−0.0220	(0.0006)			0.0683	(0.0005)
Intercept	3.3023	(0.0019)	3.3031	(0.0019)	3.0935	(0.0018)
R^2	0.8487		0.8425		0.0682	
	Based on Order-Dependent Estimates: Persons First					
Industry Average α	0.8011	(0.0019)	0.8324	(0.0017)		
Industry Average ψ	0.2410	(0.0151)			−0.6659	(0.0150)
Intercept	3.1126	(0.0019)	3.1088	(0.0018)	3.0687	(0.0019)
R^2	0.9580		0.9213		0.2486	

Notes: The dependent variable is the 84 industry-effects estimated by least squares controlling for labor force experience (through quartic), seniority, region, year, education (eight categories) and sex (fully interacted). See Table III for the regression results. The independent variables are the industry averages for the indicated firm-specific compensation policy, adjusted for the same independent variables. The time period is 1976–1987.

TABLE VIII

GENERALIZED LEAST SQUARES ESTIMATES OF THE RELATION BETWEEN FIRM-SIZE WAGE EFFECTS AND FIRM-SIZE CATEGORY AVERAGES OF FIRM-SPECIFIC COMPENSATION POLICIES

Independent Variable	Coefficient	Standard Error	Coefficient	Standard Error	Coefficient	Standard Error
	Based on Order-Independent Estimates					
Firm-Size Category Average α	1.2222	(0.0043)	1.3245	(0.0041)		
Firm-Size Category Average ψ	0.2233	(0.0026)			0.4278	(0.0025)
Intercept	3.7397	(0.0022)	3.6737	(0.0021)	3.5215	(0.0021)
R^2	0.9604		0.8960		0.2559	
	Based on Order-Dependent Estimates: Persons First					
Firm-Size Category Average α	1.1372	(0.0045)	1.3224	(0.0042)		
Firm-Size Category Average ψ	0.9217	(0.0085)			1.7395	(0.0079)
Intercept	3.6370	(0.0022)	3.6674	(0.0021)	3.3665	(0.0019)
R^2	0.9990		0.8950		0.4327	

Notes: The dependent variable is the 25 firm-size category effects estimated by least squares controlling for labor force experience (through quartic), seniority, region, year, education (eight categories) and sex (fully interacted). See Table III for full results. The independent variables are the firm-size category averages for the indicated firm-specific compensation policy, adjusted for the same independent variables. The time period is 1976–1987.

Read:

- Table VII decomposes inter-industry wage differentials into industry-average person and firm components.
- Industry-average person effects explain most inter-industry wage variation.
- Firm effects explain much less of the industry wage differential.
- Table VIII performs a similar decomposition for firm-size wage effects.
- Figure 1 shows that industry wage effects align strongly with industry-average person effects.
- Figure 2 shows a much weaker relationship with industry-average firm effects.
- Figure 3 shows that the firm-size effect is strongly related to average person effects as well as to some firm effects.

Economic message: High-wage industries and large firms pay more largely because of the workers they employ, not simply because they have high firm wage policies.

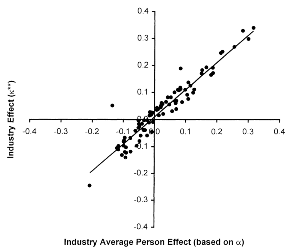

FIGURE 1.—Actual and predicted industry effects using industry-average person effects.

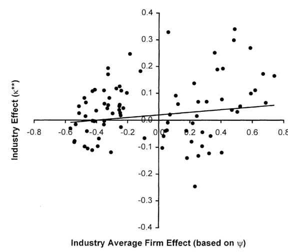

FIGURE 2.—Actual and predicted industry effects using industry-average firm effects.

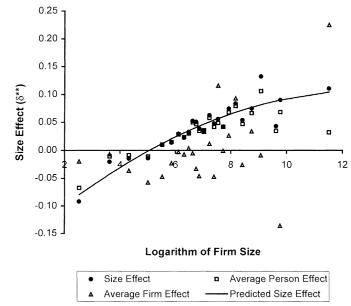

FIGURE 3.—Firm size effects related to firm-size average person and firm effects.

5.6 Tables IX–XII: firm-level outcomes

Read:

- Table IX gives summary statistics for firm-level variables and firm averages of compensation components.
- Table X relates productivity and profitability to average worker components and firm wage-policy components.
- Firms that hire high-wage workers are more productive, but not necessarily more profitable.
- Firms that pay high wages after controlling for worker effects are more productive and more profitable.
- Table XI relates compensation components to firm factor use.
- Table XII studies firm survival using proportional hazards models.

TABLE IX

SUMMARY STATISTICS FOR FIRMS
ANNUAL AVERAGES OVER ALL YEARS FOR WHICH THE FIRM DOES BUSINESS 1978–1988
(weighted by inverse sampling probability)

Variable Definition	Mean	Std Dev
Order-Independent Estimates		
Average Predicted Effect of x Variables ($x\beta$) at the Firm	0.3569	0.2586
Average Individual Effect, Unobserved Factors (α) at the Firm	−0.0575	0.6626
Average Education Effect ($u\eta$) of Employees at the Firm	3.8889	0.2757
ϕ, Firm Effect Intercept	−0.1791	1.0279
γ, Firm Effect Seniority Slope	0.0156	0.1167
Order-Dependent Estimates: Persons First		
Average Predicted Effect of x Variables ($x\beta$) at the Firm	0.3906	0.2420
Average Individual Effect, Unobserved Factors (α) at the Firm	−0.0549	0.6446
Average Education Effect ($u\eta$) of Employees at the Firm	3.8503	0.2836
ϕ, Firm Effect Intercept	−0.0196	0.2707
γ, Firm Effect Seniority Slope	0.0027	0.0775
γ_2, Firm Effect Change in Slope at 10 Years	−0.0031	0.1728
Other Firm Characteristics		
Number of Employees Sampled at Firm	34.2950	610.4800
Employment at December 31st (thousands)	0.1097	1.6789
Real Total Assets (millions FF 1980)	59.4769	3,938.9800
Operating Income/Total Assets	0.1254	0.4544
Value Added/Total Assets	1.0051	1.8889
Real Total Compensation (millions FF 1980)	1.3260	2.3570
Real Value Added/Employee (thou. FF 1980)	106.7672	936.5212
Real Total Assets/Employee (thou. FF 1980)	363.0707	21,067.5500
(Engineers, Professionals and Managers)/Employee	0.2362	0.4072
Skilled Workers/Employee	0.5414	0.5255
Log(Real Total Assets)	1.7711	3.3558
Log(Real Value Added/Employee)	4.5215	1.1050
Log(Real Sales/Employee)	5.5673	2.0139
Log(Total Employment at December 31)	−3.0262	2.1109
Log(Real Total Assets/Employee)	4.7972	2.2710
Age of Firm ($N = 7{,}385$)	19.5023	23.0331
Number of Firms	14,717	

Notes: Order-independent estimates are based upon $x\beta$, α, $u\eta$ estimated with persons first (conditional on Z) and ϕ and γ estimated with firms first (conditional on Z). Order dependent estimates are all based upon persons first ($x\beta$, α, and $u\eta$) and firms (ϕ, γ, and γ_2) second.

TABLE X

GENERALIZED LEAST SQUARES ESTIMATES OF THE RELATION BETWEEN
PRODUCTIVITY, PROFITABILITY AND COMPENSATION POLICIES

Dependent Variable:	Log (VAdded/Worker)		Log(Sales/Employee)		Operating Inc./Capital	
Independent Variable	Coefficient	Standard Error	Coefficient	Standard Error	Coefficient	Standard Error
Based on Order-Independent Estimates						
Average Predicted Effect of x Variables ($x\beta$)	0.4937	(0.0270)	0.3050	(0.0393)	0.0670	(0.0151)
Average Individual Effect (α)	0.2234	(0.0108)	0.0809	(0.0156)	0.0081	(0.0060)
Average Education Effect ($u\eta$)	0.1338	(0.0254)	−0.0057	(0.0369)	−0.0107	(0.0143)
ϕ, Firm Effect Intercept	0.0371	(0.0084)	0.0054	(0.0122)	0.0138	(0.0047)
γ, Firm Effect Seniority Slope	−0.1210	(0.0582)	−0.1751	(0.0847)	−0.0028	(0.0328)
(Engineers, Professionals, Managers)/Employee	0.3428	(0.0238)	0.1773	(0.0346)	−0.1303	(0.0126)
(Skilled Workers)/Employee	0.1226	(0.0177)	0.3065	(0.0257)	0.0061	(0.0099)
Log(Capital/Employee)	0.2470	(0.0037)	0.5536	(0.0054)		
Intercept	3.0206	(0.1055)	0.1065	(0.1533)	0.1897	(0.0579)
Based on Order-Dependent Estimates: Persons First						
Average Predicted Effect of Variables ($x\beta$)	0.6057	(0.0310)	0.4833	(0.0494)	0.0569	(0.0161)
Average Individual Effect (α)	0.2617	(0.0118)	0.1623	(0.0188)	0.0102	(0.0061)
Average Education Effect ($u\eta$)	0.0725	(0.0275)	−0.0674	(0.0437)	−0.0036	(0.0143)
ϕ, Firm Effect Intercept	0.1240	(0.0343)	0.1128	(0.0546)	0.0415	(0.0179)
γ, Firm Effect Seniority Slope	0.1492	(0.1195)	0.2852	(0.1902)	0.0571	(0.0623)
γ_2, Firm Effect Change in Slope at 10 Years	−0.0485	(0.0428)	−0.1107	(0.0681)	−0.0264	(0.0223)
(Engineers, Tech., Managers)/Employee	0.6815	(0.0247)	0.8989	(0.0394)	−0.1267	(0.0126)
(Skilled Workers)/Employee	0.2167	(0.0190)	0.4979	(0.0302)	0.0094	(0.0099)
Log(Capital/Employee)	0.1017	(0.0025)	0.2290	(0.0039)		
Intercept	4.3985	(0.1126)	2.9784	(0.1791)	0.1664	(0.0586)

Note: Models were estimated using 14,717 firms with complete data. All regressions include a set of 2-digit industry effects.

Economic message: The paper distinguishes "firms with high-wage workers" from "firms with high wage policies." This distinction matters for productivity, profitability, capital intensity, and firm survival.

TABLE XI

GENERALIZED LEAST SQUARES ESTIMATES OF THE RELATION BETWEEN FACTORS USE AND COMPENSATION POLICIES

Independent Variable	Dependent Variable: Log(Employees)	Log(Real Capital)	Log(Capital /Employee)	EPM /Employee	Skilled W /Employee	Unskilled W /Employee
Based on Order-Independent Estimates						
Average Predicted	0.2586	1.0369	0.7783	0.1420	0.0542	−0.1962
Effect of x ($x\beta$)	(0.0675)	(0.0971)	(0.0600)	(0.0110)	(0.0140)	(0.0132)
Average Individual	0.2967	0.7673	0.4705	0.1197	−0.0284	−0.0913
Effect (α)	(0.0267)	(0.0384)	(0.0237)	(0.0043)	(0.0054)	(0.0051)
Average Education	0.4380	0.5479	0.1100	0.2974	−0.1060	−0.1915
Effect ($u\eta$)	(0.0638)	(0.0918)	(0.0567)	(0.0102)	(0.0130)	(0.0123)
ϕ, Firm Effect Intercept	−0.2654	−0.2898	−0.0244	0.0315	0.0152	−0.0468
	(0.0212)	(0.0304)	(0.0188)	(0.0035)	(0.0044)	(0.0042)
γ, Firm Effect Seniority	0.4305	0.4149	−0.0156	0.0909	−0.0147	−0.0762
Slope	(0.1465)	(0.2106)	(0.1300)	(0.0241)	(0.0306)	(0.0290)
(Eng., Prof., Managers)/	−0.0479	2.0645	2.1123			
Employee	(0.0565)	(0.0812)	(0.0501)			
(Skilled Workers)/	−0.2505	0.1075	0.3580			
Employee	(0.0444)	(0.0638)	(0.0394)			
Intercept	−3.6868	2.6123	6.2991	−0.7097	0.8567	0.8530
	(0.2587)	(0.3719)	(0.2296)	(0.0420)	(0.0534)	(0.0506)
Based on Order-Dependent Estimates: Persons First						
Average Predicted	0.2541	1.0205	0.7665	0.1142	0.0628	−0.1770
Effect of x ($x\beta$)	(0.0724)	(0.1036)	(0.0638)	(0.0117)	(0.0150)	(0.0142)
Average Individual	0.2764	0.7454	0.4690	0.1231	−0.0316	−0.0914
Effect (α)	(0.0273)	(0.0391)	(0.0241)	(0.0043)	(0.0055)	(0.0052)
Average Education	0.3478	0.4076	0.0598	0.3307	−0.0964	−0.2343
Effect ($u\eta$)	(0.0643)	(0.0921)	(0.0567)	(0.0101)	(0.0129)	(0.0122)
ϕ, Firm Effect Intercept	0.3748	0.7618	0.3869	0.0057	−0.0052	−0.0005
	(0.0802)	(0.1148)	(0.0707)	(0.0131)	(0.0167)	(0.0158)
γ, Firm Effect Seniority	−0.0262	0.5277	0.5539	0.0835	−0.0303	−0.0532
Slope	(0.2798)	(0.4005)	(0.2467)	(0.0456)	(0.0582)	(0.0553)
γ_2, Firm Effect Change	0.0011	0.0497	0.0486	−0.0314	0.0140	0.0174
in Slope at 10 Years	(0.1002)	(0.1435)	(0.0884)	(0.0164)	(0.0209)	(0.0198)
(Engi., Prof., Managers)/	−0.1181	2.0038	2.1219			
Employee	(0.0568)	(0.0812)	(0.0500)			
(Skilled Workers)/	−0.2947	0.0707	0.3654			
Employee	(0.0445)	(0.0637)	(0.0392)			
Intercept	−3.4129	3.0371	6.4499	−0.8485	0.8309	1.0176
	(0.2630)	(0.3765)	(0.2319)	(0.0423)	(0.0539)	(0.0512)

Notes: The models were estimated using the 14,717 firms with complete data. All equations include a set of 2-digit industry effects. Standard errors in parentheses.

TABLE XII

PROPORTIONAL HAZARDS ESTIMATES OF THE RELATION BETWEEN FIRM SURVIVAL AND COMPENSATION POLICIES

Independent Variable	Parameter Estimate	Standard Error	Risk Ratio
Based on Order-Independent Estimates			
Average Predicted Effect of x ($x\beta$)	2.2163	(0.5821)	9.1730
Average Individual Effect (α)	−0.5874	(0.2100)	0.5560
Average Education Effect ($u\eta$)	−2.3441	(0.5327)	0.0960
ϕ, Firm Effect Intercept	0.3833	(0.1579)	1.4670
γ, Firm Effect Seniority Slope	1.2239	(1.0215)	3.4000
(Eng., Prof., Managers)/Employee	0.2328	(0.3689)	1.2620
(Skilled Workers)/Employee	0.2065	(0.2917)	1.2290
Based on Order-Dependent Estimates: Persons First			
Average Predicted Effect of x ($x\beta$)	2.0751	(0.6241)	7.9650
Average Individual Effect (α)	−0.5327	(0.2064)	0.5870
Average Education Effect ($u\eta$)	−1.8615	(0.5398)	0.1550
ϕ, Firm Effect Intercept	−0.5909	(0.5356)	0.5540
γ, Firm Effect Seniority Slope	1.6497	(2.4598)	5.2050
γ_2, Firm Effect Change in Slope at 10 Years	0.3592	(0.6677)	1.4320
(Eng., Prof., Managers)/Employee	0.4096	(0.3699)	1.5060
(Skilled Workers)/Employee	0.3372	(0.2926)	1.4010

Notes: Negative coefficients indicate a reduced probability of firm death. This model was estimated using the 7,382 firms with known birth dates. The model includes a set of 2-digit industry effects.

5.7 Appendix Table B7: basic individual-level variables

Read:

- Appendix Table B7 reports basic individual-level descriptive statistics by sex for 1976–1987.
- It includes real total annual compensation, log compensation, experience, seniority, education shares, region, and occupation categories.
- It reports the number of observations and persons in the estimation sample.

Economic message: The appendix table is a useful compact reference for the underlying wage and worker distributions used in the AKM decomposition.

DATA APPENDIX TABLE B7

DESCRIPTIVE STATISTICS FOR BASIC INDIVIDUAL LEVEL VARIABLES BY SEX FOR 1976 TO 1987

Variable Definition	Men Mean	Men Standard Deviation	Women Mean	Women Standard Deviation
Real Total Annual Compensation Cost, 1,000FF 1980	89.0967	61.6302	67.3646	37.4208
Log (Real Annual Compensation Cost, 1980 FF)	4.3442	0.5187	4.0984	0.4801
Total Labor Force Experience	17.2531	11.8258	15.4301	12.0089
(Total Labor Force Experience)^2/100	4.3752	4.9197	3.8230	4.9440
(Total Labor Force Experience)^3/1,000	13.1530	19.4305	11.6079	19.6863
(Total Labor Force Experience)^4/10,000	43.3453	77.9542	39.0589	80.3251
Seniority	7.7067	7.5510	6.5437	6.5268
Lives in Ile-de-France (Paris Metropolitan Region)	0.2561		0.2910	
No Known Degree	0.3064	0.2190	0.2971	0.2124
Completed Elementary School	0.1556	0.1458	0.1893	0.1739
Completed Junior High School	0.0565	0.0792	0.0869	0.1008
Completed High School (Baccalauréat)	0.0528	0.0804	0.0711	0.0881
Basic Vocational-Technical Degree	0.2652	0.1849	0.1926	0.1545
Advanced Vocational-Technical Degree	0.0701	0.0893	0.0532	0.0802
Technical College or University Diploma	0.0469	0.0754	0.0838	0.1247
Graduate School Diploma	0.0465	0.0964	0.0259	0.0551
Year of data	81.3106	3.7250	81.4730	3.7180
Number of Observations for the Firm in Sample	4,402.3800	16,164.6200	1,605.3100	7,797.1300
Observations	3,434,530		1,870,578	
Persons	711,518		454,787	
Proportion with Identified Least Squares Estimate of Individual and Firm Effect	0.7425		0.7448	

Source: Authors' calculations based on the Déclarations annuelles des salaires (DAS).

6 Short summary

- The paper develops one of the first large-scale matched employer–employee decompositions of wages.
- The core model decomposes compensation into observable characteristics, person effects, firm effects, and residual variation.
- Identification of firm effects relies on worker mobility across firms.
- Person effects are more important than firm effects in explaining individual wage variation.
- The unobserved person component is especially important.
- Firm effects matter, but they account for a smaller share of wage heterogeneity.
- Inter-industry wage differentials are driven mostly by worker/person effects.
- Firm-size wage premia are also driven mostly by worker/person effects.
- Firms employing high-wage workers are more productive, capital intensive, and high-skill intensive.
- Firms with high wage policies after controlling for worker quality are more productive and profitable.
- The paper created the empirical foundation for what is now called AKM worker–firm fixed-effect analysis.

7 Conclusion

7.1 Core conclusion

Abowd, Kramarz, and Margolis show that wages are shaped by both worker heterogeneity and firm wage policies, but the worker component is the dominant source of variation. High-wage workers and high-wage firms are related, but they are not the same object.

7.2 Economic intuition

A firm can appear to pay high wages for two very different reasons. It may employ workers with high person effects, or it may have a high firm wage premium. Standard wage regressions confound these channels. Matched employer–employee data allow the decomposition, and the paper finds that many classic wage differentials mostly reflect the allocation of workers across firms and industries.

7.3 Takeaway

The paper's lasting contribution is methodological and substantive. Methodologically, it made worker–firm fixed-effect decomposition central to labor economics. Substantively, it showed that person effects, firm effects, and sorting must all be considered when interpreting wage inequality, industry wage differentials, firm-size wage premia, and the link between wages and firm performance.